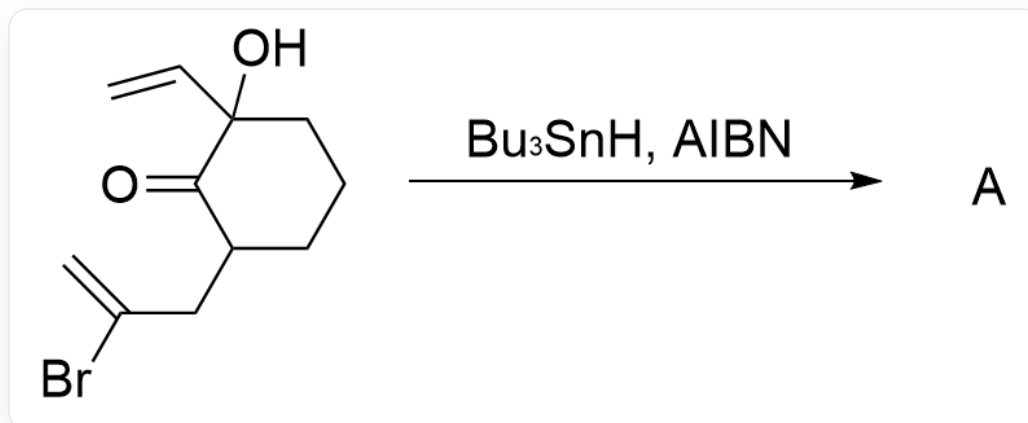


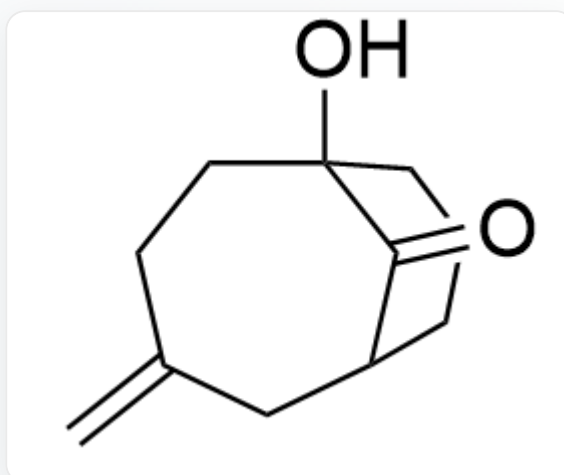
Question



O=C1C(CC(Br)=C)CCCC1(C=C)O > Bu_3SnH , AIBN > A, where Bu_3SnH is tributyltin hydride, AIBN is azobisisobutyronitrile, and the hydroxyl and hydrogen groups on the two α -positions of the carbonyl group are cis

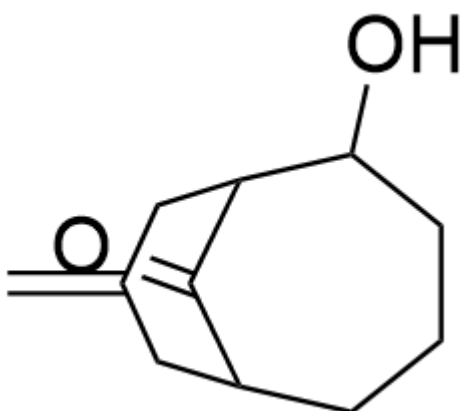
Please select the major product A of the reaction.

A.



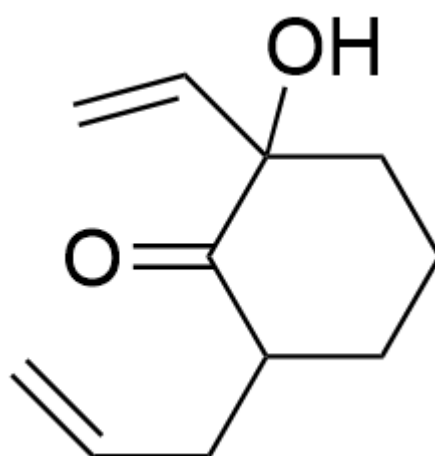
O=C1C(C2)CCCC1(O)CCC2=C

B.



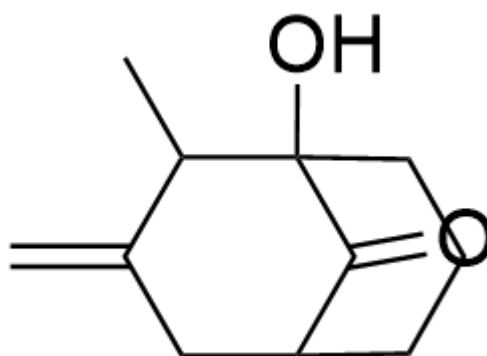
O=C1C(C2)CCCC(O)C1CC2=C

C.



O=C1C(CC=C)CCCC1(C=C)O

D.



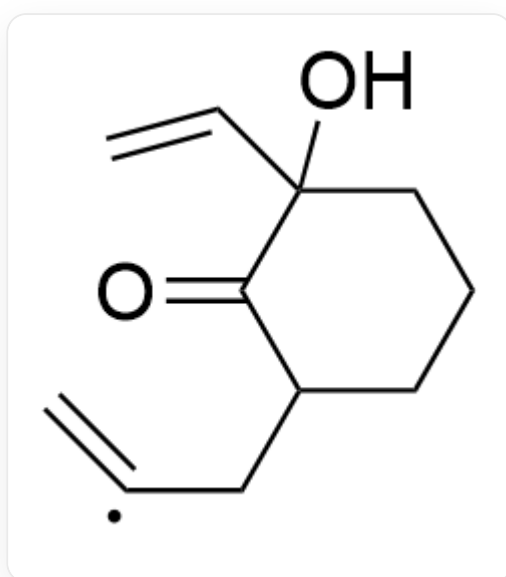
O=C1C(CC2=C)CCCC1(C2C)O

Answer

Correct Answer: **B**

Detailed Explanation

After radical initiation, an alkenyl radical is first generated.



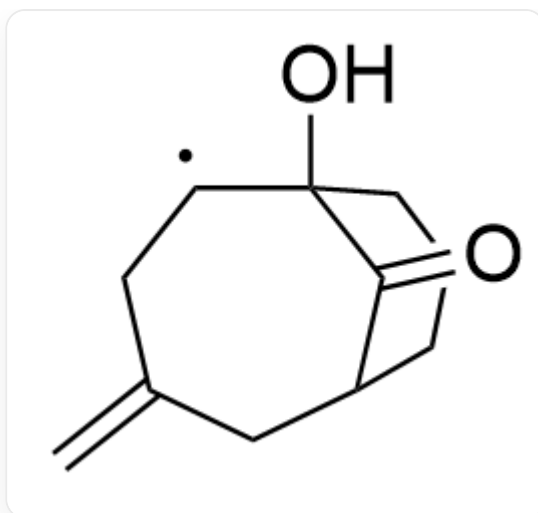
O=C1C(C[C]=C)CCCC1(C=C)O

CHECKPOINT

1 PTS

After radical initiation, an alkenyl radical O=C1C(C[C]=C)CCCC1(C=C)O is first generated

Since a secondary carbon radical is more stable than a primary carbon radical, a 7-endo-trig radical cyclization then occurs, forming a secondary carbon radical.



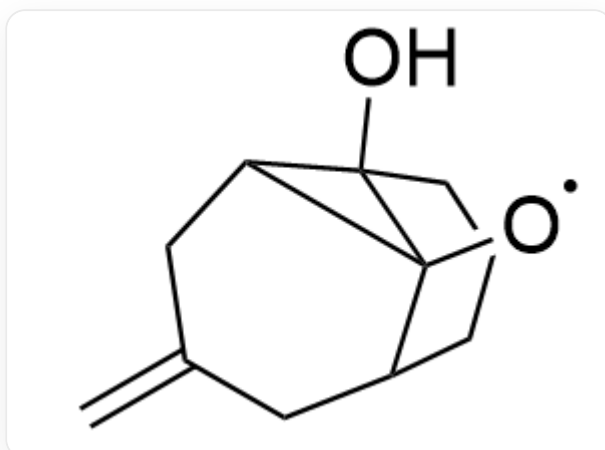
O=C1C(C2)CCCC1(O)[C]CC2=C

CHECKPOINT

1 PTS

Since a secondary carbon radical is more stable than a primary carbon radical, a 7-endo-trig radical cyclization then occurs, forming a secondary carbon radical O=C1C(C2)CCCC1(O)[C]CC2=C

This radical then attacks the carbon atom of the carbonyl group, rapidly forming an oxygen radical containing a three-membered ring.



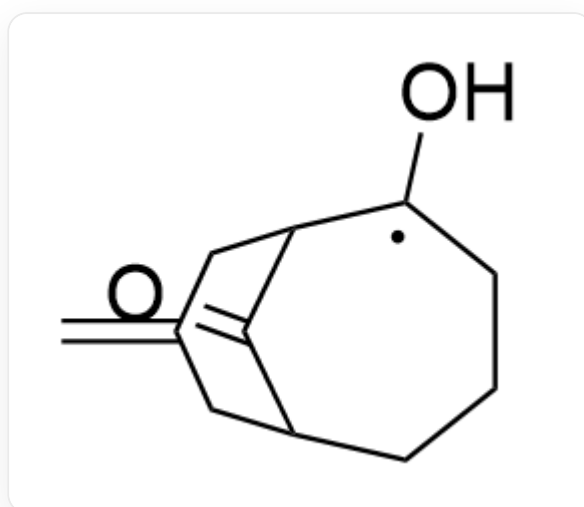
[O]C12C(C3)CCCC1(O)C2CC3=C

CHECKPOINT

1 PTS

This radical then attacks the carbon atom of the carbonyl group, rapidly forming an oxygen radical containing a three-membered ring [O]C12C(C3)CCCC1(O)C2CC3=C

The three-membered ring then undergoes cleavage, forming a more stable radical intermediate.


O=C1C(C2)CCC[C](O)C1CC2=C

CHECKPOINT

1 PTS

The three-membered ring then undergoes cleavage, forming a more stable radical intermediate O=C1C(C2)CCC[C](O)C1CC2=C

Finally, this radical is quenched by *Bu₃SnH* to obtain the final product A.

CHECKPOINT

1 PTS

Finally, this radical is quenched by Bu_3SnH